Lab 7 Fuzz Testing
Spam Scoring Filter

SRN: PES2UG24CS036
Date: 8 November 2026

Assigned repository: https://github.com/SETAPESU26/L6_L7_33_spam_scoring

Source: lab7 | Baseline commit: 985be01d0936e946070685493cea685739decbfb

Result

All three input-validation functions are fixed. The six original discovery checks now pass. The expanded suite passes 72 tests, including 13 Hypothesis properties with 500 successful examples each: 6,500 successful generated examples in the final full run.

Failures found before the fixes

Function	Fuzz input	Original failure	Fixed result
sender_domain	None	AttributeError	empty string
sender_domain	empty string	IndexError	empty string
first_link	None	AttributeError	empty string
first_link	empty string	ValueError	empty string
shout_ratio	None	TypeError	0.0
shout_ratio	empty string	ZeroDivisionError	0.0

Method and submission mapping

The original application was executed, six discovery properties were written and failed against the original source, then the application was hardened and the same checks rerun. Additional properties and regression examples verify useful outputs as well as no-crash behavior.

The handout names processor.py and test_processor.py generically. This assigned repository uses spam_scoring.py and test_spam_scoring.py; those filenames are preserved.

Screenshots show real captured subprocess output displayed in Chromium. Logs and exit-code metadata are retained. The final full-suite screenshot is labeled as an excerpt; its complete output and statistics are included in evidence/logs/.

Step 2 Run the original application

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 01_before_normal | Exit code: 0
Command: python spam_scoring.py
Working directory: /home/adityashrivastav/SE_Lab/Lab-7/original
stdin (escaped): "offers@Deals.com\nVisit https://example.com/deal now\nFREE prize NOW\n"
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): Sender Domain: deals.com

==== Running first_link ====
Enter a message containing a link: First Link: https://example.com/deal

==== Running shout_ratio ====
Enter a message (e.g. FREE prize NOW): Uppercase Ratio: 0.5833333333333334
```

A normal run succeeds: deals.com; https://example.com/deal; ratio 7/12. All three supplied inputs are recorded in the screenshot header.

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 02_before_invalid | Exit code: 1
Command: python spam_scoring.py
Working directory: /home/adityashrivastav/SE_Lab/Lab-7/original
stdin (escaped): "\n\n\n"
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): Traceback (most recent call last):
  File "/home/adityashrivastav/SE_Lab/Lab-7/original/spam_scoring.py", line 33, in <module>
    main()
    ~~~~^^
  File "/home/adityashrivastav/SE_Lab/Lab-7/original/spam_scoring.py", line 19, in main
    result = sender_domain(user_input)
  File "/home/adityashrivastav/SE_Lab/Lab-7/original/spam_scoring.py", line 2, in sender_domain
    return email.split('@')[1].lower()
           ~~~~~~~~~~~~~~~~^^^
IndexError: list index out of range
```

The same original program exits with code 1 on an empty email address. The traceback ends with IndexError, before the later prompts can run.

Steps 3 and 4 Mixed input fuzzing fails

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 03_before_fuzz | Exit code: 1
Command: python -m pytest test_discovery.py -q -k not text_no_crash --tb=short --color=no
Working directory: /home/adityashrivastav/SE_Lab/Lab-7/original
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
FFF                                                                      [100%]
================================== FAILURES ===================================
_____________________________ test_sender_domain_no_crash _____________________________
test_discovery.py:14: in test_sender_domain_no_crash
    @given(INPUTS)

test_discovery.py:16: in test_sender_domain_no_crash
    result = sender_domain(value)
             ^^^^^^^^^^^^^^^^^^^^
spam_scoring.py:2: in sender_domain
    return email.split('@')[1].lower()
           ^^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'split'
E   Failing test case: test_sender_domain_no_crash(
E       value=None,
E   )
_____________________________ test_first_link_no_crash _____________________________
test_discovery.py:22: in test_first_link_no_crash
    @given(INPUTS)

test_discovery.py:24: in test_first_link_no_crash
    result = first_link(value)
             ^^^^^^^^^^^^^^^^^
spam_scoring.py:5: in first_link
    start = text.index('http')
            ^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'index'
E   Failing test case: test_first_link_no_crash(
E       value=None,
E   )
_____________________________ test_shout_ratio_no_crash _____________________________
test_discovery.py:30: in test_shout_ratio_no_crash
    @given(INPUTS)

test_discovery.py:32: in test_shout_ratio_no_crash
    result = shout_ratio(value)
             ^^^^^^^^^^^^^^^^^^
spam_scoring.py:12: in shout_ratio
    letters = [c for c in text if c.isalpha()]
                          ^^^^
E   TypeError: 'NoneType' object is not iterable
E   Failing test case: test_shout_ratio_no_crash(
E       value=None,
E   )
=========================== short test summary info ============================
FAILED test_discovery.py::test_sender_domain_no_crash - AttributeError: 'None...
FAILED test_discovery.py::test_first_link_no_crash - AttributeError: 'NoneTyp...
FAILED test_discovery.py::test_shout_ratio_no_crash - TypeError: 'NoneType' o...
3 failed, 3 deselected in 0.16s
```

Hypothesis minimizes each mixed-type failure to None: sender_domain raises AttributeError, first_link raises AttributeError, and shout_ratio raises TypeError. Three tests fail; three text-only tests are deselected.

Steps 3 and 4 Text-only fuzzing fails

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 03b_before_text_fuzz | Exit code: 1
Command: python -m pytest test_discovery.py -q -k text_no_crash --tb=short --color=no
Working directory: /home/adityashrivastav/SE_Lab/Lab-7/original
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
FFF                                                                      [100%]
================================== FAILURES ===================================
_____________________ test_sender_domain_text_no_crash ______________________
test_discovery.py:38: in test_sender_domain_text_no_crash
    @given(st.text())
                   ^^^
test_discovery.py:40: in test_sender_domain_text_no_crash
    assert isinstance(sender_domain(value), str)
                      ^^^^^^^^^^^^^^^^^^^^
spam_scoring.py:2: in sender_domain
    return email.split('@')[1].lower()
           ^^^^^^^^^^^^^^^^^^^
E   IndexError: list index out of range
E   Failing test case: test_sender_domain_text_no_crash(
E       value='',
E   )
_____________________ test_first_link_text_no_crash _____________________
test_discovery.py:44: in test_first_link_text_no_crash
    @given(st.text())
                   ^^^
test_discovery.py:46: in test_first_link_text_no_crash
    assert isinstance(first_link(value), str)
                      ^^^^^^^^^^^^^^^^^
spam_scoring.py:5: in first_link
    start = text.index('http')
            ^^^^^^^^^^^^^^^^^^
E   ValueError: substring not found
E   Failing test case: test_first_link_text_no_crash(
E       value='',  # or any other generated value
E   )
_____________________ test_shout_ratio_text_no_crash _____________________
test_discovery.py:50: in test_shout_ratio_text_no_crash
    @given(st.text())
                   ^^^
test_discovery.py:52: in test_shout_ratio_text_no_crash
    assert 0.0 <= shout_ratio(value) <= 1.0
                 ^^^^^^^^^^^^^^^^^^^
spam_scoring.py:14: in shout_ratio
    return len(upper) / len(letters)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
E   ZeroDivisionError: division by zero
E   Failing test case: test_shout_ratio_text_no_crash(
E       value='',
E   )
=========================== short test summary info ===========================
FAILED test_discovery.py::test_sender_domain_text_no_crash - IndexError: list...
FAILED test_discovery.py::test_first_link_text_no_crash - ValueError: substri...
FAILED test_discovery.py::test_shout_ratio_text_no_crash - ZeroDivisionError:...
3 failed, 3 deselected in 0.17s
```

Text-only strategies independently minimize all three inputs to an empty string. They reveal IndexError, ValueError, and ZeroDivisionError respectively. These are genuine failed runs against the preserved original application.

Step 5 Run the fixed application

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 04_after_normal | Exit code: 0
Command: python spam_scoring.py
Working directory: /home/adityashrivastav/SE_Lab/Lab-7
stdin (escaped): "offers@Deals.com\nVisit https://example.com/deal now\nFREE prize NOW\n"
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): Sender Domain: deals.com

==== Running first_link ====
Enter a message containing a link: First Link: https://example.com/deal

==== Running shout_ratio ====
Enter a message (e.g. FREE prize NOW): Uppercase Ratio: 0.5833333333333334
```

The normal demonstration is unchanged. Its complete output is byte-for-byte identical to the original normal-run log.

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 05_after_invalid | Exit code: 0
Command: python spam_scoring.py
Working directory: /home/adityashrivastav/SE_Lab/Lab-7
stdin (escaped): "\n\n\n"
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): Sender Domain:

==== Running first_link ====
Enter a message containing a link: First Link:

==== Running shout_ratio ====
Enter a message (e.g. FREE prize NOW): Uppercase Ratio: 0.0
```

Three blank inputs now finish with exit code 0. Sender domain and first link are empty strings; the uppercase ratio is 0.0. The application reaches all three prompts.

Input handling

Non-string inputs return safe empty/zero results. Addresses are checked before splitting; link searches tolerate missing or malformed URLs; the ratio checks its denominator before dividing.

Step 6 Re-run the fuzz tests

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 06_after_discovery | Exit code: 0
Command: python -m pytest test_spam_scoring.py -k no_crash -q --tb=short --color=no
Working directory: /home/adityashrivastav/SE_Lab/Lab-7
Complete stdout and stderr from the recorded subprocess, displayed verbatim below.

```
......                                                                   [100%]
6 passed, 66 deselected in 0.90s
```

All six original discovery properties pass against the fixed module; 66 additional cases are deselected for this direct comparison.

Lab 7 - Captured command output

SRN: PES2UG24CS036 | 07_after_full_suite | Exit code: 0
Command: python -m pytest test_spam_scoring.py -v --tb=short --color=no --hypothesis-show-statistics
Working directory: /home/adityashrivastav/SE_Lab/Lab-7
Output excerpt: final 14 lines. Complete output is preserved in evidence/logs/07_after_full_suite.txt

```
  - Stopped because settings.max_examples=500


test_spam_scoring.py::test_nonletters_do_not_change_shout_ratio:

  - during generate phase (0.14 seconds):
    - Typical runtimes: < 1ms, of which < 1ms in data generation
    - 500 passing, 0 failing, and 0 invalid test cases

  - Stopped because settings.max_examples=500


============================== 72 passed in 1.89s ==============================
```

The final full suite passes 72 tests with exit code 0. This screenshot is explicitly the final 14 lines of the verbose output. All 13 properties report 500 passing generated examples in the complete text log.

Coverage beyond crash discovery

The final suite includes 59 deterministic regression cases and 13 generated properties. It checks correct domain extraction, invalid addresses, first-of-two link selection, Unicode whitespace, quoted links, punctuation, IPv6, malformed ports, Unicode case counting, numeric/container inputs, and 100,000-character inputs.

A long unmatched-bracket URL test checks the linear trimming implementation. No test is skipped or marked as an expected failure. The fixed default pytest configuration excludes the intentionally broken original artifacts.

Implementation and reproduction

sender_domain

Accept one simple email-like address with nonempty parts. Strip outer whitespace, reject internal whitespace/control characters and malformed labels, and lowercase the domain. This helper is not a full RFC mailbox validator and performs no network checks.

first_link

Find HTTP(S) schemes case-insensitively; stop at whitespace, quotes or angle brackets. Remove prose punctuation while preserving balanced brackets, check the hostname/port, skip malformed candidates, and return the first acceptable URL.

shout_ratio

Count Unicode alphabetic characters and uppercase alphabetic characters in one pass. Return uppercase/letters when letters exist; otherwise return 0.0. Nonletters do not change the ratio and all outputs are in [0.0, 1.0].

Commands from Lab-7

```
python -m pip install -r requirements.txt
python spam_scoring.py
python -m pytest test_spam_scoring.py -v
python tools/run_evidence.py before
python tools/run_evidence.py after
```

Environment: Python 3.13.15; Hypothesis 6.170.15; pytest 9.2.0. The pinned test dependencies require Python 3.11 or later. The application has no external runtime dependencies.

Required deliverables

This screenshot PDF; spam_scoring.py; test_spam_scoring.py. Original files, test configuration, pinned dependencies, logs, screenshots and reproduction tools are also included.

Submission: https://github.com/Audatic07/SE_Lab/tree/main/Lab-7

Reference: Hypothesis strategies and @given/settings API documentation at https://hypothesis.readthedocs.io/en/latest/reference/strategies.html and https://hypothesis.readthedocs.io/en/latest/reference/api.html